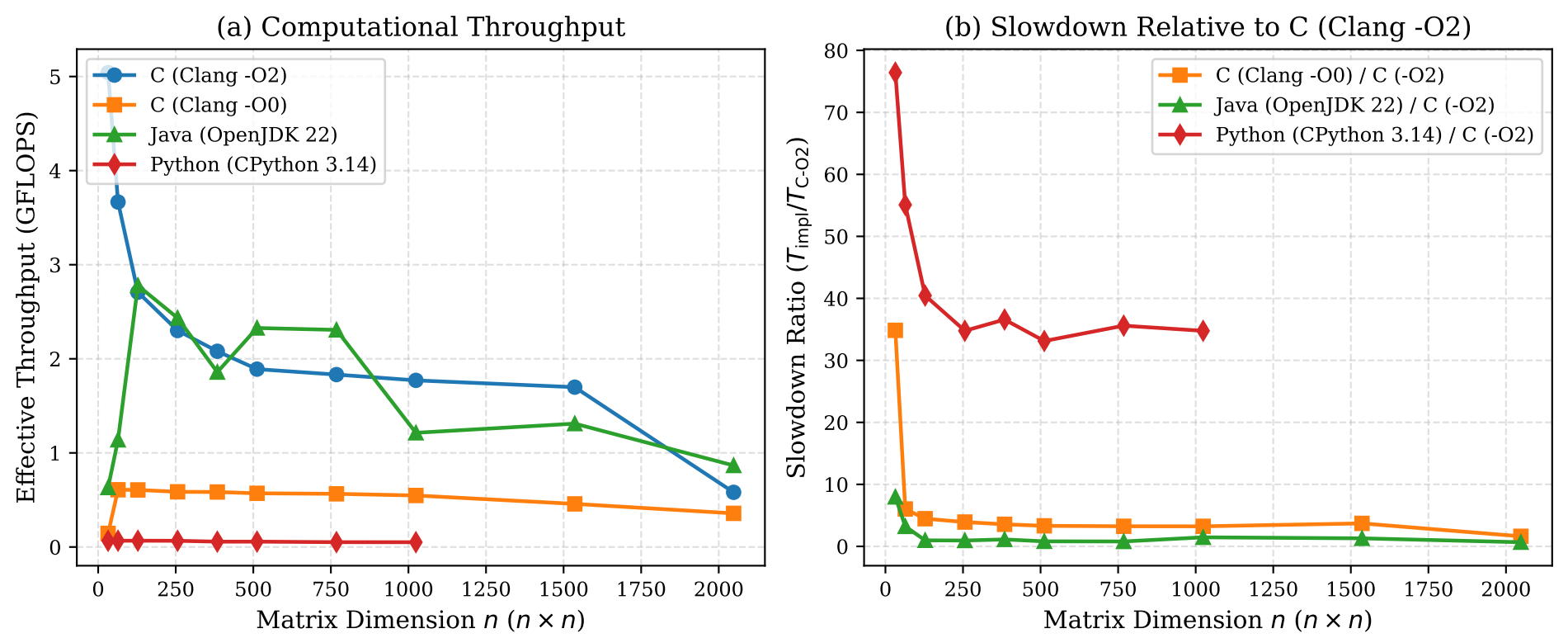
(a) Computational Throughput
Effective Throughput (GFLOPS)
Matrix Dimension n ($n \times n$)
C (Clang -O2)
C (Clang -O0)
Java (OpenJDK 22)
Python (CPython 3.14)
0
1
2
3
4
5
0
250
500
750
1000
1250
1500
1750
2000
(b) Slowdown Relative to C (Clang -O2)
Slowdown Ratio ($T_{\text{impl}}/T_{\text{C-O2}}$)
Matrix Dimension n ($n \times n$)
C (Clang -O0) / C (-O2)
Java (OpenJDK 22) / C (-O2)
Python (CPython 3.14) / C (-O2)
0
10
20
30
40
50
60
70
80
0
250
500
750
1000
1250
1500
1750
2000